Input: A rectangle $R^{\text{root}} = X_1 \times X_2 \times \cdots \times X_k$, where each X_i equals $[N]^{cN^{1-\alpha_i}}$.

Output: A rectangle $R^{\text{cur}} = X_1^{\text{cur}} \times X_2^{\text{cur}} \times \cdots \times X_k^{\text{cur}}$, and k sets $J_1, J_2, \cdots, J_k$.

for each i, $J_i \leftarrow [cN^{1-\alpha_i}]$;

$R^{\text{cur}} \leftarrow R^{\text{root}}$;

while *R^{cur} is not in a leaf level* [a] **do**

  without loss of generality, we assume it is player i's turn to speak;

  X_i^{cur} is partitioned by: $X_i^{\text{cur}} = X^0 \cup X^1$, and R^{cur} is thus partitioned by: $R^{\text{cur}} = R^0 \cup R^1$;

  toss a $(\frac{|X^0|}{|X_i^{\text{cur}}|}, \frac{|X^1|}{|X_i^{\text{cur}}|})$ biased coin c;

  if $c = 0$:

    $X_i^{\text{cur}} \leftarrow X^0$;

    $R^{\text{cur}} \leftarrow R^0$;

  if $c = 1$:

    $X_i^{\text{cur}} \leftarrow X^1$;

    $R^{\text{cur}} \leftarrow R^1$;

  if $X_i^{\text{cur}}(J_i)$ is $(1 - \frac{1}{10k \log N})$-dense:

    continue;

  else:

    decompose X_i^{cur} by Lemma 13 with $J = J_i$, get $X^1, \cdots, X^r, I_1, \cdots, I_r$;

    R^{cur} is thus decomposed by $R^{\text{cur}} = R^1 \cup \cdots \cup R^r$;

    sample a random element $\boldsymbol{j} \in [r]$: $\boldsymbol{j}$ w.p. $|X^j|/|X_i^{\text{cur}}|$ equals j for each j;

    $X_i^{\text{cur}} \leftarrow X^j$, $J_i \leftarrow J_i \backslash I_j$;

    $R^{\text{cur}} \leftarrow X_1^{\text{cur}} \times X_2^{\text{cur}} \times \cdots \times X_k^{\text{cur}}$;

[a] R^{cur} is not in a leaf level means R^{cur} is not a sub-rectangle of any leaf rectangle of the protocol tree.